Aziz's Notes - 9618 Paper 1

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"# NOTES MADE FROM PASTPAPERS AND MARKSCHEMES (the stuff that actually shows up on the exam)" - Aziz, 2025

Chapter 1: Information Representation

Number Systems:

- **Converting from binary to denary:** Sum the individual position values, understanding that the least significant bit represents 2^0.
- **Converting between binary and hexadecimal:**
  - From binary to hexadecimal: Start with the four least significant bits and convert them to one hexadecimal digit, proceeding upwards towards the most significant bit, successively taking groups of four bits and converting each group to the corresponding hexadecimal digit.
  - From hexadecimal to binary: Treat each digit separately, converting it into a 4-bit binary equivalent (e.g., F converts to 1111, E converts to 1110).

One's and Two's Complement

- **One’s complement:** Each binary digit in a binary number is individually subtracted from 1 which means that each 0 is switched to 1 and each 1 is switched to 0.
- **Two’s complement:** The binary number obtained if 1 is added to the one’s complement number.
- The leftmost bit in the binary value in a two’s complement number is **0** if the number is positive and **1** if the number is negative. This bit remains unchanged through the process of converting a binary number to two’s complement.
- **Maximum and minimum of an 8-bit two’s complement number:** The maximum is +127 and the minimum is -128.
- An 8-bit unsigned binary integer can represent 256 different values.

Binary Coded Decimal (BCD)

- **Definition:**

> It uses four bits (nibble) to represent a denary digit. Each four bits can only represent a denary digit smaller than or equal to 9.

- **Conversion:** Each denary digit is written as a 4-bit binary number.
- **Applications:**
  - Electronic devices like calculator displays and digital clocks.
  - Financial applications requiring accurate decimal value representation.
  - Date and time storage in BIOS of PCs or game consoles.
  - Barcodes.
- **Advantages:**
  - Straightforward conversion from BCD to denary.
  - Easier for digital equipment to display output information.
  - Exact representation of monetary values.

Internal Coding of Text

Character Set

- **Definition:**

> All of the characters/symbols that the computer recognizes/uses. Each character has a unique binary number. It is a list of characters recognized by the computer hardware and software.

ASCII vs. Unicode

- **Similarities:**
  - Both can use 8 bits.
  - Both represent each character using a unique code.
  - Unicode will contain all the characters that ASCII contains/ASCII is a subset of Unicode.
- **Differences:*

Feature	ASCII	Unicode
Range of Characters	Limited	Greater range, representing most written languages.
Language Support	Primarily English	Multiple languages

Feature	ASCII	Unicode
Bit Usage	7 or 8 bits (1 byte)	2 bytes (up to 4 bytes)
Standardization	Not standardized	Standardized
Relationship	Subset of Unicode	Superset of ASCII

- **How computers represent characters using ASCII codes:**
  - Each character has a unique character code (binary number). Each character in the word is replaced by its code. The binary value for each character in the string is stored in the order of the string/word. ASCII codes may be 7 or 8 bits long (extended ASCII coding is 8 bits long while the standard ASCII coding is 7 bits long).

Images

Vector Graphics

- **Representation:** A series of instructions about how to draw (geometric) shapes. Each item in the image is a drawing object. It contains a drawing list which contains commands for creating each drawing object and the properties of that object(data about the shapes) such as the fill colour, line colour, width of line, shape, position, dimensions/size, radius (in case it is a circle)…etc. Objects are created using mathematical calculations and the calculation for proportional size is stored (exact dimensions are not stored).

Key Image Terms

- **Pixel:**

  > A single square of one color / The smallest addressable element in an image

- **Image Resolution:**

  > The number of pixels in the image OR the product of the number of pixels per row times the number of rows

- **Screen Resolution:**

  > The number of pixels which can be viewed horizontally and vertically on the screen.

- **Bit Depth:**

> number of bits per pixel/used to represent each colour.

- **Drawing List:**

> a list that stores the data required to draw each drawing object in the graphic.

- **Drawing Object:**

> a component created using a formula.

- **Property:**

> it defines one characteristic of a component such as the line thickness or the fill colour.

- **Impact of Changing Bit Depth:** Increasing the bit depth means that the image has a greater range of colours. Increasing the bit depth makes the image closer to the original.
- **Impact of Increasing Image Resolution:** More pixels can be stored. The image is less pixelated (sharper).

Bitmap Images

- **Definition:**

> An image is made up of pixels. Each pixel is of a single color. Each color has a unique binary number. Stores the sequence of binary numbers for each image / frame by storing the binary value for each pixel. It is suitable for photographs / scanned images. It uses less processing power than vector graphics. It can be compressed with significant reduction in file size unlike vector graphics images which don't compress well because of little redundant data.

- **Bitmap File Storage:** A bitmap file stores pixel data values, pixel dimensions, file header, and resolution.
- **File Header:** Stores data about the bitmap image such as:
  - File size in bytes and dimensions/size of the image.
  - Colour/bit depth/number of bits per pixel.
  - Resolution of the image (horizontal width of image in pixels, vertical height of image in pixels).
  - Type of compression used.
  - File type (confirmation that it is a bitmap)
- **Advantages of Vector Graphics over Bitmap Images:**
  - Can be enlarged without becoming pixelated as the instructions to create the image are stored/it is recalculated. This is useful when displaying images on different sized screens.

- Smaller file size as it stores coordinates and calculations/instructions about the shapes, not individual pixels → faster transfer/downloading rates and reduced storage requirements when stored many times.
- Easier in editing than bitmaps, because commands can be easily adjusted unlike bitmap where each pixel would need to be edited separately.

Color Calculation

- The number of possible colors is calculated
  by 2number of bits for each pixel=maximum number of colors2number of bits for each pixel =maximum number of colors.

Graphics Software Features

- **Resize:** Increase/decrease size of the image or object in the image.
- **Color select:** Select all pixels of the same color.
- **Add text.**
- **Fill an area with color.**
- **Copy:** Replicate a number of pixels.
- **Select:** Grab a number of pixels to perform a task to/with.

Sound

Analogue Data

- **Definition:**

> Data values that are continuously changing.

Encoding Sound

- **Sampling:** The amplitude of the sound wave is recorded at regular intervals. The value of each amplitude is given a corresponding binary number. The binary numbers of the samples are stored in sequence.
- **Sampling Rate:**

> It is the number of samples taken per unit time. Increasing sampling rate means that samples will be closer together, so the digital sound waveform is closer to the original analogue waveform and more accurate due to smaller quantization errors. However,

> more samples will be recorded per second, so more binary values will be stored per second, increasing file size, and increasing downloading time.

- **Sampling Resolution:**

> It is the number of bits used to encode each sample. It is referred to as bit depth. Increasing sampling resolution increases the number of bits per sample → each binary amplitude is closer to the analogue amplitude and the digital soundwave is closer to the original analogue soundwave → more accurate representation of the analogue sound → reduces quantization errors→ greater range of values can be stored→ crisper sound quality. However, more bits will be stored per sample, so more data is stored, increasing file size and increasing downloading time.

Sound Editing Software Features

- **Amplify:** Increase the volume of a section of sound.
- **Change pitch:** Increase/decrease frequency of section of sound.
- **Change sampling resolution:** to change the accuracy of the sound/ file size.
- **Cut/delete:** Remove part of the sound file.

MP3 Music Quality

- MP3 uses perceptual music shaping, removing noises that may not be heard by the human ear. If two sounds are played at the same time, the softer sound is removed. This is done to give a closer representation of the original track without affecting the quality too much.

Compression Techniques

Lossy vs. Lossless Compression

Feature	Lossy Compression	Lossless Compression
Data Integrity	Data is lost; decompressed file is not the same as the original.	No data is lost; original file can be recovered.
Use Cases	High compression ratio is needed, and when it is not important for data to be maintained or originals to be recovered.	When all the original data needs to be retained, usually in text files and program source codes.

Reasons for Compressing Files

- Reduce the storage space required to store the file/ Reduce the file size.
- Reduce the time needed to transmit/download the file.

Bitmap Images and Email Compression

- Reduced bandwidth usage and time when transmitting email.
- Reduced storage space on the email.
- Email accounts often have a maximum size for an attachment.

Benefits of Photograph Compression for Customers

- Customers will download the photographs in less time → photos take less of the customer's bandwidth.
- Photographs take less space on the customer's storage medium.
- Customers can store more images.

Use Cases: Lossy vs Lossless

- **Lossy:** Higher compression ratio for images, little identifiable change to the viewed images, loss of quality won't be noticed.
- **Lossless:** The images may be required in their original form/ may need to be recovered completely. The image may already have a small file size so major file size reduction is not needed as the files will load quickly.

Lossy vs Lossless Compression in Compressing a Real-Time Video

- **Lossy Compression:** Loss of quality won't be noticed. Smaller file sizes will reduce buffering so the video can be played more smoothly. Needs to be viewed in real-time so less bandwidth is needed if the file size is smaller.
- **Lossless Compression:** Original recording may not have been made in high resolution. Could be streaming to high bandwidth devices. Viewers don't want any loss of quality.

Image and Sound Compression (Lossy)

- **Images:**
  - Reduce the color depth by reducing the number of bits per color.
  - Reduce the resolution by reducing the number of pixels per unit measurement.
- **Sound:**
  - Reduce sampling rate by reducing the number of samples taken per second.
  - Reduce sample resolution by reducing the number of bits used to represent each sample.

Run-Length Encoding (RLE)

- **Definition:**

> It is a lossless method of compression. It encodes a repeating string of adjacent, identical characters/pixels into two values: the number of times the character/pixel is repeated and the character code/ colour code of the pixel.

- **Text Files:**
  - Huffman coding: analysis to identify the most often used characters and given shorter character codes.
  - Run-length encoding: Instead of storing each repeated sequences of characters individually, the character is stored once with the number of consecutive occurrences.

Why Lossy Compression Should Not Be Used on a Text File

- None of the original data can be lost.
- The text file would be corrupted/the text file cannot be opened.

RLE Limitations

- RLE stores a color and the number of times it occurs consecutively.
- An image may not have many sequences of the same color.
- It would need to store each color and count 1 which adds data.
- Example: Red Green Blue would be Red 1 Green 1 Blue 1.

How RLE Compresses Images

- Replace sequences of the same color pixels with color code once and the number of identical pixels.

How RLE Compresses Sound Files

- Reduce the amplitude to only the range used → limited amplitudes mean fewer bits per sample.
- For consecutive sounds that are the same, the binary value of the sound is recorded along with the number of times it is repeated. Only the changes are recorded instead of actual sounds.

RLE for Video Frames

- The frame is made up of pixels, where each pixel has a binary value for its colour. Consecutive pixels with the same binary value/colour are stored as a single data value

along with their count.

Overflow

- **Definition:**

> It is a problem that can occur in binary addition. It is a condition when the result of a calculation is too large to fit into the number of bits defined for storage. It can cause the computer to interpret the result as a negative number although it is a positive number and vice versa.

Chapter 2: Communication and Networking Technologies

Client-Server Model

- **Description:**

> The server stores the files/data/database. The computers used to request data are the clients. A client sends requests to the server. The server searches for the data needed and returns the results of the request made to the client.

- **Examples of applications:**
  - Sending and receiving email.
  - A company or school centrally storing files.
  - Using a print server.
  - Using a file server.
  - Self-checkout machines at supermarkets.
  - Customers accessing their information at a bank.
- **Accessing a file on a website:** The file is made available from a web server. The user's browser is the client software. The client software browser requests the file from the server. The desired file is returned to the client computer.
- **Benefits:**
  - Files and resources are centralized.
  - Creation of security.
  - Centralized back-up.
  - Internet monitoring.
  - Clients can be less powerful machines, so less expensive to buy.
  - Saving resources on the server reduces the burden on the client.

- **Thin-client model:** Server performs all processes required by the task and/or data storage. Clients only send requests to the server and display the returned results.
- **Thick-client model:** The server performs minimal processing for the client. The clients also do most of their own processing and work independently.

File Sharing: Peer-to-Peer Networks

- **Features:**
  - All computers are of equal status.
  - Each computer provides access to resources and data.
  - Computers can communicate and share resources.
  - Each computer is responsible for its own security.
- **Drawbacks:**
  - Reduced security (no central management of security) as each computer is at risk from viruses from other computers.
  - No central management of backup so if the data from one computer is not backed up it is lost to all of them.
  - No central management of files so consistency may be difficult to maintain because each computer may have different software from the others.

Transmission Media

- **Types of Connections:**
  - **Fibre-optic:** Transmits data as light.
  - **Satellite:** A communication device in Earth's orbit that receives and transmits data.
  - **Radio waves:** Carries data wirelessly in the form of electromagnetic waves.
  - **Copper cable:** Carries data as electrical signals and can consist of a twisted pair.
- **Fiber-Optic Cables:**
  - **Benefits:** Less (noise) interference in the signal. The signal does not degrade as quickly so it needs less signal boosting. More secure as it is difficult to hack.
  - **Drawbacks:** Initial installation cost is higher. Specialists are needed to install and maintain. Fibers can break when bent. Only transmits data in one direction.
- **Copper Cables:**
  - **Benefits:** Less expensive to install. Easier to install because it is more flexible. Easier to make terminations.
- **Wireless vs. Wired Networks:**
  - **Wireless Benefits:** Less hardware is required so there is a reduced cost of setting up the network. Users and computers can be mobile as the network can be accessed from anywhere within range of an access point. It is much more straightforward and

easier to connect additional devices as there is no need to physically connect each device.
    - **Wireless Drawbacks:** Transmissions may be less secure as connection by unauthorized user is easier. Bandwidth may be limited as more devices connect the bandwidth so access may be slow → less performance than wired connections. Interference is more significant which can hinder transmission or corrupt data.
- **Dedicated Line vs. Internet:**
    - **Dedicated Line Benefits:** Faster communication. More stable connection and transmission speed. Improved security.
    - **Dedicated Line Drawbacks:** Expensive to set-up. Disruption to the dedicated line would leave no alternative.
- **Satellite Communication:**
    - **Benefits:** Not fixed to a single location. Allows access in remote / rural areas.
    - **Drawbacks:** High latency (slow to connect). More expensive than wired methods, as need extra equipment. Signal is affected by bad weather. The transmission speed is slower than fixed line broadband . Direct line of sight needed.

LAN (Local Area Network)

- **Benefits of Connecting Computers to a LAN:**
    - Allows sharing of files and data.
    - Allows central management such as backup/security central management.
    - Allows the sharing of resources.
- **Characteristics of a LAN:**
    - Devices connected over a small geographical area.
    - Uses dedicated infrastructure as the infrastructure is privately owned.
- **LAN vs. WAN (Wide Area Network):**

Feature	LAN	WAN
Geographical Area	Small	Large
Ownership	Private	Private or Public
Data Transfer Rate	Higher	Lower
Security	More secure, easier to implement protection	Less secure

Mesh Topology

- **Description:**

> All computers are connected to all other devices. There are multiple routes between devices. The computers can act as relays, passing packets on towards the final destination. Each end-system has a point-to-point connection to each of the other end-systems. Transmission is duplex. Messages can be unicast, multicast, or broadcast.

- **Advantages of Mesh over Bus Topology:**
  - If one line goes down, there are more routes available.
  - Improved security as not using one main line and data is sent over a dedicated connection.
  - Fewer collisions.
  - New nodes can be added without interfering with other nodes.

Star Topology

- **Advantages of Star over Bus Topology:**
  - More resilient to faults as there is no single cable, which leads to less disruption.
  - Higher performance as fewer collisions because each device is only connected to the switch (central device).
  - Easier to add new nodes because each device connects directly to the switch (central device).
- **Characteristics:**
  - Devices are connected to a central device (Switch, Hub, Router).
- **Packet Transmission:**
  - **Star Network:** Packet has address of recipient. Sender Transmits packet to switch/central device. Switch directly transmits packet to recipient only.
  - **Star Network with a Server:** Packet has address of recipient. Sender transmits packets directly to the server. Server reads address and identifies where recipient is. Server transmits packets directly to the recipient. Server transmits packets only to the recipient.
- **Hardware Devices in a LAN:**
  - Router
  - Switch
  - Hub

CSMA/CD (Carrier-Sense Multiple Access with Collision Detection)

- **How it handles collisions:**
  - Workstations monitor the communication channel.
  - Data is only sent when the channel is free.
  - Collision caused when 2 devices transmit at the same time.

- If a collision is detected, the workstations send a jamming signal.
- If a collision occurs, each workstation waits a random time before retransmitting.
- Each time a collision occurs, random time is increased.
- If no activity is detected, start transmission and continuously check for a collision.
- If no collision is detected, continue transmission.

Wireless Network Interface Card

- **Functions:**
  - Provide Interface to a wireless network as an antenna.
  - Receives analog radio waves and converts them to digital/binary.
  - Checks incoming transmissions for correct IP/MAC address and ignores transmissions not meant for it.

Firewall

- **Definition:**

> A Firewall is a device that monitors and controls incoming and outgoing network traffic based on set criteria.

Subnetting

- **Benefits of dividing a network into subnetworks by subnetting the LAN:**
  - Improves network security so that not all devices can access all segments of the network.
  - Reduces congestion as data stays in its segment so it does not travel as far.
  - Easier network management because faults can be isolated more efficiently.
  - Easier to expand the network by allowing for greater range of IP addresses to be available.
  - Improves network performance.

Internet Infrastructure

- **Communication Systems Supporting the Internet:**
  - PSTN (Public Switched Telephone Company).
  - Cell phone network: Send data to cell towers over mobile connection.
  - Dedicated lines: permanent connection used for businesses.
  - Satellite System: Send data to satellites in orbit.

- **Data Transmission Using a PSTN:** The PSTN consists of many different types of communication lines. Data is transmitted in both directions at the same time (full duplex data transmission). The communication passes through different switching centers.

Internet Infrastructure

Routers and Gateways

- **Router:** Connects devices to the internet.
  - Stores **IP** and **MAC addresses** of attached devices.
  - Receives packets, analyzes the destination **IP address**, and forwards packets.
  - Operates between similar networks (same protocol).
  - Can segment a network.
- **Gateway:** Connects a server using a different protocol to the internet.
  - Joins two different types of networks (**dissimilar networks**).

Both routers and gateways:

- Can connect two or more networks.
- Can connect a network to a **WAN**.
- Receive packets, analyze destination **IP address**, and forward using the routing table.
- Assign private **IP addresses**.

World Wide Web (WWW)

- **Internet:** Infrastructure and a global collection of networks (**interconnected network of computer devices**).
- **World Wide Web:** A collection of web pages stored on websites.
- Access to the WWW is over the internet.
- Web pages are written in **HTML**.
- **HTTP** protocol is used to transfer web pages.
- Internet uses **IP protocol**.

Cloud Computing

Accessing a service/files/software on a remote server.

- **Public Cloud:** Computing services offered by a 3rd party provider over the public Internet. Open to anyone with credentials.

- **Private Cloud:** Only available to select users, not the general public. A dedicated system accessible only to an organization.

Advantages

- Cloud storage can be free for small quantities.
- No need for separate high-capacity storage devices.
- Access data from any computer with internet access.
- Most cloud data services have built-in backup or disaster recovery.

Disadvantages

- Requires a reliable internet connection for access.
- Can take a long time to upload/download data.
- Can be more expensive long-term due to provider charges.
- There could be a limit to the amount of storage unless paid for.
- Possible loss of control due to reliance on an external agency for backup and security.

Bit Streaming

- **Bit Streaming:** A sequence of bits sent over a communication path (Internet) at high speed.
  - Requires fast broadband and buffering.
  - Bits arrive in the same order as sent.

Video Transmission

1. Data is **compressed** before transmitting.
2. Video is transmitted continuously as a series of bits.
3. Video is hosted on a **media server**.
4. On download, the server sends data to a **buffer** on the client computer.
5. Recipient receives bit stream from the buffer.

Playback without Pausing

- User needs high-speed broadband.
- Data is streamed to a buffer in the computer.
- Buffering stops video pausing as bits are streamed.
- As buffer empties, it refills, ensuring continuous viewing.
- Playback is a few seconds behind the time the data is received.

On-Demand vs. Real-Time

- **On-demand:**
  - Streaming of an event that has taken place in the past.
  - Existing media are encoded to bit streaming format and uploaded to a dedicated server.
  - A link for the encoded video is placed on a website.
  - A user clicks on the link to download the encoded streaming video and the streamed video is broadcast to the user.
  - Can be paused.
- **Real-time:**
  - A live stream of an event that is currently taking place.
  - The event is captured live with a video camera connected to a computer.
  - The video signal is converted to streaming media files encoded on the computer.
  - The encoded feed is uploaded from the computer to a dedicated streaming server.
  - The server then sends the live images to all users requesting it as real-time video streaming.
  - Cannot be paused.

Benefits

- No need to wait for the entire file to download.
- No need to store large files on the computer.
- No specialist software is required for playback in a browser.

Drawbacks

- Video stops if the internet is very slow.
- Video stops if buffering capacity is inadequate.
- Loss of internet stops the user from accessing films.

IP Addressing :LiBatteryWarning:

- **IP Address Purpose:** Gives each device on a network an identifier so it can be used to locate the device.
  - Each address is unique within the network.
  - Allows a device or node to send data to the correct destination.
- **IP Address Components:**
  - **Network ID:** Identifies the network. Each subnetwork has a different network ID.
  - **Host ID:** Uniquely identifies the device within the subnetwork.

IPv4 vs. IPv6

Feature	IPv4	IPv6
Format	4 groups of digits	8 groups of digits
Notation	Denary	Hexadecimal
Range	0-255 per group	0-FFFF per group
Separator	Full-stop	Colon
Length	32 bits (4 bytes)	128 bits (16 bytes)
Zero Handling	N/A	First instance of 0's is represented using double colons
Example Address	`192.2.1.0`	`59FB::1005:CC57:6571`

Why Use IPv6?

- The number of addresses needed will exceed the number available using IPv4.

Private vs. Public IP Addresses

Feature	Private IP Address	Public IP Address
Scope	Only known within the LAN; used for internal LAN communication	Unique throughout the internet; allows direct access on the internet
Internet Access	Requires NAT to access the internet directly	Direct access
Security	More secure	Less secure
Assignment	Assigned by the router	Provided by ISP
Uniqueness	Unique within the LAN but can be duplicated within other networks	Unique throughout the internet

Router IP Addresses

- Router needs a **public IP address** to be identified on the Internet.
- Router needs a **private IP address** to be identified on the home network.
- Laptop has only a **private IP address** because it does not connect directly to the internet, enhancing security.

Static vs. Dynamic IP Addresses

- **Static:**
  - Does not change each time the device rejoins the network.
  - Address is assigned by network/ISP.
- **Dynamic:**
  - A new one is reallocated each time a device rejoins a network.
  - Address is assigned by the network OS.
- **Servers** would use static IP addresses to prevent DNS update delays and "address not found" errors.

Domain Names

- **IP Address:** Physical address of the server.
- **Domain Name:** A memorable form for an IP address.
- Each domain name is linked to an IP address.
- A **Domain Name Server (DNS)** translates a domain name into its corresponding IP address.
- Domain name does not change, but the IP address could change if dynamic.
- IP address can be used to address the server directly.
- The domain name needs translation before the server can be accessed.

URL Conversion to IP Address

1. The browser parses the **URL** to obtain the domain name.
2. The domain name is looked up in the locally cached list of corresponding **IP addresses**.
3. If not found, the domain name is sent to the closest **DNS**.
4. The DNS stores a table of domain names and corresponding IP addresses.
5. If the DNS finds the domain name, it returns the **IP address**.
6. If it cannot find the domain name, it sends the request to a higher-level DNS.
7. If it is not found again, the request is passed to a higher-level server.
8. If the domain name is finally not found, an error message is generated.
9. The browser uses the **IP address** returned to request the required web page from the web server.
10. The web server retrieves the page and delivers it to the originator.
11. The browser software interprets the script and displays the web page.

URL Symbols

- `%20` - Coding for a space (since <space> not allowed in a URL).
- `?` - Separates the URL from all parameters or variables.

LANs and WANs

Wide Area Network (WAN)

> A network connecting computers on different sites, possibly thousands of kilometers apart.

- Used by an organization or company to connect sites or branches.
- Not owned by the organization or company (leased from a public switched telephone network company (PSTN)).
- A dedicated communication link will be provided by the PSTN.
- The transmission medium will be fibre-optic cable.
- Transmission within the WAN will be from switch to switch.
- A switch will connect the WAN to each site.
- There will not be any end-systems connected directly to the WAN.

Local Area Network (LAN)

> A network connecting computers in a single room, in a single building, or on a single site.

- Used by an organization or company within a site or branch.
- Owned by the organization or company.
- It will be one of many individual LANs at one site.
- The transmission medium will be twisted pair cable or WiFi.
- The LAN will contain a device that allows connection to other networks.
- There will be end-systems connected, which will be user systems or servers.

Benefits of a LAN

- Application software can be installed on an application server attached to the LAN instead of on each individual PC.
- A file server can be attached to the LAN to allow users to store larger files and share files.
- Printers can be attached to a print server connected to the LAN, providing higher-quality printing.
- Managers can use electronic mail to communicate with staff.
- The paper-less office is possible, with files stored in digital form on a file server.

File Sharing

- **Peer-to-Peer Model Advantages:**
  - Avoids congestion when many clients download files simultaneously.
  - Parts of a file can be downloaded separately.
  - The parts are available from more than one host.

Networking Topologies

Requirements for Data Communication

- Sender
- Receiver
- Transmission medium
- Message
- Protocol

Data Flow Modes

- **Simplex:** Flow is one way.
- **Half duplex:** Flow is 2 ways but one-way at a time.
- **Full duplex:** Flow is 2 ways and can be at the same time.

Message Types

- **Broadcasting:** 1 to all
- **Multicasting:** 1 to many
- **Unicasting:** 1 to 1

Networking Topologies

- **Point to point topology:** Two end-systems are connected by a network link.
- **Bus topology:** Has one shared link for multiple end-systems.
  - No direct connection between end-systems; messages are broadcast.
  - A fault in one end-system or the link does not affect other end-systems.
- **Star topology:** Each end-system has a point-to-point connection to a central device.
  - Transmission is duplex.
  - Messages can be unicast, multicast, or broadcast.
  - The failure of an end-system or its link does not affect other end-systems.

- The central device must not fail.

Definitions

- **Topology:**

  > The configuration of a network that defines how the various devices on the network are connected.

- **Bus topology:**

  > Contains one shared link to which all devices are attached.

- **End-system:**

  > A computer or server connected to a network.

- **Mesh topology:**

  > Contains direct links between devices.

- **Star topology:**

  > Each end-system is linked to a central device.

- **Hybrid network:**

  > A collection of connected LANs where some of them have different topologies or supporting technologies.

- **Cable:**

  > A transmission using copper wire or fibre-optic.

- **Bandwidth:**

  > A measure of the amount of data that can be transmitted per second.

Cables

	Twisted pair	Coaxial	Fibre-optic
Cost	Lowest	Higher	Highest
Bandwidth	Lowest	Higher	Much higher
Attenuation	Affected	Most affected	Least affected

	Twisted pair	Coaxial	Fibre-optic
Interference	Worst affected	Less affected	Least affected
Need for repeaters	More often	More often	Less often

LAN Hardware

- **Server:**

  > A system providing a service to end-systems.

- **Repeater:**

  > A device that connects two cables and provides full-strength signal to the second cable.

- **Bridge:**

  > A device that connects two segments of a LAN with the same protocol.

- **Network Interface Card (NIC):**

  > A component used to identify the end-system.

- **Switch:**

  > A connecting device that allows two or more devices to communicate with each other.

- **Wireless Access Point (WAP):**

  > A connecting device that allows the connection of devices using WiFi.

- **Wireless Network Interface Card (WNIC):**

  > Provides the NIC function in a WiFi LAN; a hardware component that allows a device to connect to a wireless network.

Ethernet

- A protocol for data transmission over standard wired network connections.
- Uses Carrier Sense Multiple Access with Collision Detection for handling collisions.
- Data is transmitted in frames, where each frame has a source and destination IP/MAC address.
- Error checking for data occurs, so damaged frames can be resent.

ISP Contribution

- Access ISPs give internet access to individuals or companies.
- Access ISPs connect with middle-tier or regional ISPs, which then connect to tier 1 ISPs or backbone ISPs.
- Connections between ISPs are handled by an Internet Exchange Point (IXPs).
- Other networks which share the top of the hierarchy with tier 1 ISPs are the major content providers.

PSTN Connectivity Services

- Broadband network connection for traditional network access
- WiFi hotspot technology equips a public place or area with an access point connected to a wired network for internet access.

Cloud Approaches for Organizations

1. The organization takes full responsibility for creating and managing the cloud installed on-site and connected to a private network.
2. The organization outsources to a third party the creation and management of an on-site installation connected to a private network.
3. The organization outsources the creation and management of an internet-accessible system by a third party.

Public Cloud Details

- Created, managed, and owned by a third-party cloud service provider.
- Accessible via a browser from any suitable device in any location.
- Services provided:
  - Infrastructure provision
  - Platform provision
  - Software provision

Media Streaming Data Flow

- **Media Server to Buffer:**
  - Limited by the bandwidth of the network connection.
  - If bitrate is not met by the user, different levels of compression are available.
  - The most highly compressed version will be with the poorest quality.
- **Buffer to User:**

- Must match defined rate for specific media in use, but the planned transmission rate to the buffer should be higher to allow for unexpected delays.
- Rates are controlled by a media player by monitoring high and low watermarks.
- Essential to have a buffer that doesn't get filled.

IP Addressing Schemes

- Network bits (**netID**) are used to route to the specified network.
- Host ID is used to route to the specific host/device inside this network.

Classful Approach

Class	Identifier	NetID	HostID	Starting IP	Ending IP
Class A	0	7 bits	24 bits	`0.0.0.0`	`127.255.255.255`
Class B	10	14 bits	16 bits	`128.0.0.0`	`191.255.255.255`
Class C	110	21 bits	8 bits	`192.0.0.0`	`223.255.255.255`

Classless Inter-Domain Routing (CIDR)

- Removes rigid structure and allows splitting between **netID** and **hostID** to be more flexible.
- Avoids waste of IP addresses.
- Format: `$196.27.5.21/23$` means 23 bits for **netID** and 9 bits for **hostID**.

Subnetting Approach

- Allows further structure in addressing.
- Example: Using a Class C netID with 3 bits for the network and 5 bits for hosts.

Domain Name Lookup Outcomes

1. Domain name under jurisdiction of the server, so IP is returned directly.
2. IP is cached, so IP is returned directly.
3. Domain is remote, so it goes to the root server, then to the TLD server, then to the lower domain, and then the IP is returned.

Chapter 3

Embedded Systems

Features

- Microprocessor/microcontroller within a larger system that performs one specific task.
- Usually integrated or built into a larger device or system.
- Must have a processor, memory, and input/output.
- Combination of hardware and software designed for a specific function.
- Not easily changed or updated by the owner.

Drawbacks

- Difficult to update the firmware by the user.
- Cannot be easily adapted for another task.
- Troubleshooting faults is a specialist task.

Benefits of EEPROM

- No additional equipment is needed to change firmware, enabling updates by non-technical users.
- Can be erased and reprogrammed multiple times, allowing erasure of a particular byte or the whole EEPROM.
- Possible to reprogram or update without removing it from the device.

Why ROM Is Used

- To store data that does not change.
- Data must be stored even when the device is without power.
- To store boot-up instructions.

Memory Components

Primary Storage

- Needed to store files to boot the system.
- Needed to store parts of the operating system.
- Needed to store any intermediate data to be used.

Secondary Storage

- Needed to store files or software long term.

RAM vs. ROM

Feature	RAM	ROM
Volatility	Volatile	Non-volatile
Data Alteration	Can alter data	Data can’t be altered
Read/Write	Read/write	Read only
Storage	Files/data currently in use	BIOS
Memory Size	Often larger	Smaller

Purpose of RAM

- Stores current progress of a device, e.g., printer.
- Stores currently running parts of files/programs/software.
- Stores data about the device, e.g., toner levels in a printer, data read from sensors in a washing machine.

Purpose of ROM

- Stores the bootstrap program (BIOS).
- Stores the start-up instructions for the system/device.
- Stores the kernel of the operating system for the device.

DRAM vs. SRAM

Feature	DRAM	SRAM
Refresh	Has to be refreshed frequently	Does not require a refresh
Components	Single transistor and capacitor	More than one transistor
Transistors	N/A	Transistors arranged as flip-flops or latches
Bit Storage	Each bit as a charge	Each bit using a flip-flop
Power Consumption	Higher power consumption under low levels of access	Less power
Cost	Less expensive to purchase	More expensive to purchase
Access Time	Slower access time	Faster access times
Storage/Data Density	Higher storage/data density	Lower storage/data density

Feature	DRAM	SRAM
Usage	Main memory	Cache memory

PROM, EPROM, and EEPROM

Feature	PROM	EPROM	EEPROM
Overwriting	Can be set once	Can be overwritten multiple times	Can be overwritten multiple times
Removal	N/A	Needs to be removed from device	Can be erased in situ
Erasing Method	N/A	Erased using UV light	Erased using voltage and is flash storage
Erasing Granularity	N/A	Must be entirely erased before rewriting	Does not have to be entirely erased before rewriting

Reasons for Using EEPROM in VR Headset

- EEPROM allows multiple read/write operations, meaning that the contents of the firmware in the headset can be changed easily without fully erasing its contents first.

Buffers

> Whenever data has to be transferred from one part of a computer system to another, a problem occurs if the data can be sent more quickly than it can be received. The solution to the problem is to use a buffer.

- Data enters a buffer before being transmitted to its destination.
- The buffer functions as a queue, so the data emerges in the order that it has entered the buffer.
- Typically, the buffer is created in the computer memory.

Secondary Storage Devices

Hard Disk Drive Operation

- The hard disk has one or more platters made of aluminum or glass.
- Each surface of the platter is ferrous-oxide, which is capable of being magnetized.
- The platters are mounted on a central spindle.

- The entire mechanism is contained inside a sealed (aluminum) box.
- The disks are rotated at high speed.
- Each surface of the disk has a read/write head mounted on an arm positioned just above the surface.
- A cushion of air ensures that a head does not touch the platter surface.
- Electronic circuits control the movement of the arm and, hence, the heads.
- The surface of the platter is divided into concentric tracks and sectors.
- One track in one sector is the basic unit of storage called a block.
- The data is encoded as a magnetic pattern for each block.
- When writing to disk, a variation in the current in the head produces a variation in the magnetic field on the disk.
- When reading from disk, a variation in the magnetic field produces a variation in the current through the head.

Optical Disc Reader/Writer

How does the CD-ROM operate?

- Spins the disc.
- Tracking mechanism moves the laser assembly.
- A lens focuses the laser onto the disc.
- Laser beam is shone onto disc to read / write.
- Surface of disc has a reflective metal layer / phase change metal alloy.
- Tracks on the disc have sequence of pits and lands / amorphous and crystalline state.
- Reflected light is then encoded as a bit pattern.
- **CD-ROM** makes use of **pits and lands** when reading data.
- **CD-RW and DVD-RW** make use of **amorphous and crystalline states** (special alloy material) when reading and writing data.
- When writing data to the disc, the laser light shone changes the material to either crystalline or amorphous depending on its intensity.
- When reading data from the disc, laser light is reflected from the crystalline solid but not from the amorphous allowing the coding of 1 or 0.

How does the DVD-RAM operate?

- Data is written using lasers.
- **DVD-RAM** uses **phase changing recording**, in which varying laser intensities cause targeted areas in the phase change recording layer to alternate between an amorphous and

a crystalline state.
- It uses a rotating disk with concentric tracks.
- It allows read and write operations to occur simultaneously.

DVD-RW vs DVD-RAM

Feature	DVD-RW	DVD-RAM
Track Type	Single spiral track	Several concentric tracks
Read/Write Operations	Read OR write as separate operations	Allows simultaneous read/write operations
Software Requirements	Requires special packet reading/writing software	Requires no special read/write software
Data Writing	Existing data must be completely erased to write new data	Makes use of sectors to store data
Disc Rotation Speed	Disc rotates at different speeds	Disc rotates at a constant speed

Solid State Memory

- Has **no moving parts**.
- It is **non-volatile**.
- It makes use of **arrays of semiconductors** or **NAND gates**.
- It uses grids of columns and rows that has two transistors at each intersection.
- One transistor is called a **floating gate** and the second is called a **control gate**.
- The movement of electrons controls read/write operations.
- It is necessary to first erase old data in order to write new data in the same location.
- An example of solid state memory technology is **flash memory** which is mostly **NAND-based**.

Secondary Storage Devices

Examples:

- Internal/External SSD
- Internal/External HDD
- Pen drive

External/Removable Secondary Storage Devices

Used for:

- Additional secondary file storage
- Backup of files
- Archiving of files
- Transfer files to second computer

SSD vs HDD

Feature	SSD	HDD
Access Speed	Fast access speed	Slower
Reliability	Reliable as it has no moving parts, so it is not easily damaged	More susceptible to damage due to moving parts
Noise	Quiet as it has no moving parts	Can be noisy due to moving parts
Data Degradation	Faster degradation of data	Slower degradation of data, so more longevity
Capacity	Large capacity	Large storage capacity
Cost	More expensive per storage unit	Inexpensive per storage unit

Magnetic Storage vs Solid State Storage

Magnetic storage is more appropriate than solid state storage for a computer used to store videos recorded by a CCTV system because:

- The computer will have a large number of read/write operations because it is working all the time, and magnetic storage is more suitable as it has more longevity.
- Magnetic storage costs less per storage unit, which is important as videos are large files and therefore very large storage capacity is required.

Screens and Associated Technologies

Resistive Touch Screen

- Resistive screens consist of two charged plates with an air gap between the layers.
- When a finger touches the screen, the pressure causes the plates to touch as the top layer moves to touch the bottom layer, completing the circuit.
- The point of contact is registered.
- Coordinates are used to calculate the position.

- A processor determines the horizontal and vertical coordinates of the point of contact.

Capacitive Touch Screen

- Capacitive screen is made from materials that store electric charge.
- When touched, the charge is changed and drawn to the point of contact and transferred to the finger.
- Sensors at the screen corners detect the change in the electrostatic field.
- Coordinates of contact are registered.
- Coordinates are used to calculate the position (horizontal and vertical position is calculated).

Ports

- HDMI port
- VGA port
- USB port

Laser Printer

- The revolving drum is initially given an electrical charge.
- A laser beam bounces off moving mirrors scans back and forth across the drum discharging certain points.
- The drum is coated with oppositely charged toner which only sticks to charged areas.
- The drum rolls over electro-statically charged paper.
- The pattern on the drum is transferred to the paper.
- The paper is passed through the fuser to seal the image.
- The electrical charge is removed from the drum and the excess toner is collected.

3D Printer

- The digital 3D model is designed using Computer Aided Design (CAD) software.
- The software splits the object into slices as the model is built one layer at a time starting from the bottom.
- The data about the slices is sent to the printer.
- The solid plastic is heated and pushed through the nozzle.
- A stepper motor moves the nozzle into position.
- The nozzle forces out the molten plastic.
- This is all repeated until the layer is complete.

- Each subsequent layer is completed through the same process.
- The material is fused together layer by layer.

Temperature Sensor in a 3D Printer

- Purpose: To prevent overheating by identifying the temperature of the object being printed and the material being used.

Peripheral Device Ports for Printers

- USB port
- COM port

Input and Output of Sound

Microphone

- The microphone has a diaphragm.
- The incoming sound waves cause vibrations of the diaphragm.
- Causing:
  - A coil to move past a magnet (dynamic microphone).
  - Changing the capacitance (condenser microphone).
  - Deforms the crystal (crystal microphone).
- An electrical signal is produced.

Speaker

- **Components**
- Diaphragm
- Coil of wire
- Permanent magnet

How it works

- An electrical signal is taken and translated into physical vibrations to create sound waves through the following steps.
- An electric current in the coil creates an electro-magnetic field.
- Changes in the audio signal causes the direction of the electric current to change.
- The direction of the current determines the polarity of the electro-magnet.

- The electro-magnet is repelled by or attracted to the permanent magnet repeatedly, causing the coil to vibrate.
- The movement of the coil causes the diaphragm to vibrate.
- The vibration is transmitted to the air in front of the diaphragm as sound waves.
- The amount of movement will determine the frequency and amplitude of the sound wave produced.

Virtual Reality Headset

- Most important components of a virtual reality headset are the two eye-pieces.
- Two eye-pieces are fed paired images from the controlling system → when looked at together, give the eyes the sensation of being in a 3D environment.
- Images can be collected using specialised photographic techniques or can be created using a 3D graphics package.
- The wearer of the headset can control which part of the 3D environment is in view by moving their head or by using a controlling device.

Chapter 5: Computer Architecture

Von Neumann Model

Stored Program Concept

> The stored program is a sequence of instructions and data that are stored in the same memory space/main memory.

- The processor has direct access to memory.
- The memory also contains the data required by the program.
- Each instruction is fetched, decoded, and executed by the processor.
- The processor executes instructions sequentially.
- Instruction fetch and data operation cannot occur at the same time.
- The processor uses several **registers** to store the data and instructions from the program because they can be accessed faster than main memory.

Central Processing Unit Architecture

ALU and CU Functions

- Arithmetic and Logic Unit (ALU):
  - Performs arithmetic operations.
  - Performs logical comparisons.

- Control Unit (CU):
  - Sends and receives control signals along the control bus to control operations and communication between CPU components.
  - Synchronises operations of other components in the CPU.
  - Manage the execution of instructions in sequence.
  - Transmits signals to coordinate events based on the pulses of the system clock.

System Clock

- Generates timing signals which are sent on the control bus.
- Generates signals to synchronise events in the processor or fetch-decode-execute cycle and other system components.
- Synchronize operations and make sure processes are operated in sequence.

Clock Speed

- The clock sends out a number of pulses in a given time interval, where the clock pulses synchronize the components on the motherboard.
- Each instruction is executed on a clock pulse → clock speed dictates the number of instructions that can be run per second.
- Higher clock speed means more fetch-execute cycles per second/ more instructions can be run per second.
- However, there is a limit on the clock speed because the heat generated by higher clock speeds cannot be removed fast enough.

Number of Cores

- Each core processes one instruction per clock pulse.
- Multiple cores mean that sequences of instructions can be split between them.
- So more than one instruction is executed per clock pulse (more sequences of instructions can be run at the same time).
- More cores decrease the time taken to complete tasks.

Multiple Cores vs Single Core

Explain why the second computer doesn’t always run twice as fast as the first computer given that the first has one 2.1 GHz CPU while the second computer has two 2.1 GHz cores.

- Multiple cores introduce additional overheads because of the need for communication between cores.
- The two computers may have more differences than just the cores as one may have more RAM → allowing faster multitasking. Also, one may have a GPU.

- Software may not be designed for multiple cores → One of the cores remains unused.

Immediate Access Store (IAS)

- IAS stores all the data currently in use.
- IAS is a volatile memory.
- IAS has fast access time.

Cache Memory

- The fastest component of the IAS.
- Performance improves with increased storage size for the cache and with increased rate of access.
- The higher the capacity of the cache, the more frequently used instructions it can store for fast access → less time is spent accessing from RAM.
- Fastest access is obtained by having all or part of the cache on the CPU chip.

Registers

What is a Register?

- It is a small piece of memory.
- It is part of the processor.
- It is temporary storage of data that is about to be or has been processed.

Accumulator

- A general-purpose register that stores a value before and after the execution of an instruction by the ALU.

Special-Purpose Registers

- Program counter: Stores the address of the next instruction to be fetched.
- Memory address register (MAR): Holds the address where data/instruction is currently being read from or saved to/fetched.
- Memory data register (MDR): Holds data that is about to be written to memory or has just been read from memory / stores the data from the address pointed to by the MAR. The data in it is copied to the CIR.
- Current instruction register: Holds the instruction that is currently being decoded/executed
- Status register:

    - It is interpreted as independent bits or flags.
    - It stores flags which are set depending on events.
    - Example: Addition overflow/ result of operation is zero.
- Index register: Stores a value that is added to an address to give another address.

Special-Purpose vs General-Purpose Registers

eature	Special Purpose Registers	General Purpose Registers
Function	Hold the status of a program	Hold the temporary data while performing operations
Specialization	Specialized for a specific use	Used for any purpose
Instruction Usage	Can only be used by certain instructions	Can be used by most instructions

System Bus

Bus Usage

- Address bus:
    - Carries the address where data is being written to or read from.
    - Receives data from the MAR.
    - Carries data from registers to main memory.
    - Uni-directional (one direction).
- Data bus:
    - Carries data between processor or registers and memory.
    - Carries data that is currently being processed.
    - Carries an address or an instruction or a value.
    - Bidirectional (two directions).
- Control bus:
    - Transmits signals between the control unit and the other components (main memory).
    - Transmits timing signals to components.
    - Bidirectional.

Control Bus Signals

- Interrupt signals
- Timing signals
- Read/Write signals

Data Bus Width

- The width of the data bus determines the number of bits that can be simultaneously transferred.
- Increasing the width of the data bus increases the number of bits that can be moved at one time.
- This improves the processing speed as fewer transfers are needed.

Address Bus Width

- Increasing the number of directly addressed memory locations.

Optical Disc Reader/Writer

Describe the roles of the address bus, the data bus, and buffers in the process of writing the optical disc reader/writer.

- A buffer temporarily holds data until it is ready to be transmitted to the device.
- The address bus of the data to be written to the device is carried on the address bus.
- All data to be written to the device/buffer is carried on the data bus.

Ports

Purpose of a Port

- Provide the connection to peripheral devices.
- Provide an interface between the computer and other I/O devices.

Examples:

- USB
- HDMI

USB Ports

- Devices are automatically detected and configured when first attached (plug and play).
- It is nearly impossible to wrongly connect a device.
- USB has become an industrial standard.
- It is supported by many operating systems.
- Later versions are backward compatible with earlier USB systems.
- Allows power to be drawn to charge portable devices.

Data Transmission through a USB Port

- One bit is transferred at a time.
- It can be synchronous or asynchronous.
- USB-3 is full duplex and earlier versions are half-duplex.

What is an Interrupt?

> A signal sent from a source/device telling the processor that its attention is needed.

Examples

- Hardware:
  - Printer out of paper.
  - No CD in drive.
- Software:
  - A running program needs input.
  - Runtime error such as division by zero.
  - Attempt to access an invalid memory location.

How Interrupts are Handled

- At the start or end of each fetch-execute cycle the processor checks for interrupts
- If an interrupt flag is set in the interrupt register, the processor identifies source of interrupt.
- If interrupt is high enough/higher than current process, then lower priority interrupts ae disabled and the processor saves the current contents of registers on stack.
- Processor calls appropriate interrupt handler or Interrupt Service Routine (ISR) to handle interrupt and the address of ISR is loaded into program counter (PC).
- When ISR is finished, processor checks for further interrupts.
- If no further interrupts, the processor restores the contents of registers from stack back to the registers and the processor continues with the next Fetch-Execute cycle.

Fetch-Decode-Execute Cycle

Stages

- The Program Counter (PC) holds the address of the next instruction to be fetched.
- The address in the Program Counter (PC) is copied to the Memory Address Register (MAR) using the address bus.
- The contents of Program counter (PC) are incremented.
- The data/instruction is held in the Memory Data Register (MDR) from the address held in the MAR using the data bus.
- The instruction from the MDR is copied to the Current Instruction Register (CIR)

- The processor's instruction set is used to decode the instruction into op code and operand. Then, the processor executes the instruction.

Register Transfer Notation

- MAR←[PC]. The contents of the PC are copied to the MAR
- MDR←[[MAR]]. The data in the location pointed to by the MAR is copied to the MDR.
- PC←[PC]+1. The address in PC is incremented. Both this step and the previous one take place simultaneously.
- CIR←[MDR]. The contents of MDR are copied into CIR.

Double Brackets

- The contents of the MAR is an address, it is the contents of that address which is transferred to MDR
- The contents of the address pointed to by the MAR is transferred to the MDR

Chapter 6: Assembly Language Programming

Assembly Language

Macro

> A group of instructions is given a name. A group of instructions that need to be executed several times within the same program.

- The statements are written once and called using the name whenever they need to be executed.
- Macro code is inserted into the source file at each place it is called.

Directive

> An instruction that directs the assembler to do something. A directive is not a program instruction.

- It is information for the assembler.
- Examples are:
  - State the start address for the program.
  - Tell the assembler to set aside space for variables.
  - Include an external file.

Symbolic, Relative, and Absolute Addressing

Symbolic Addressing

> The operand is a word/symbol that represents the memory location/address.

- Example is: ADD Num1

Absolute Addressing

> The operand is a numeric address.

- Example is: ADD 230

Relative Addressing

> A number (offset) is added to the base address to give the actual address.

Assembler's First Pass

On the first pass of the two-pass process, the assembler adds entries to a symbol table. Explain how the assembler makes these entries (first pass process).

- The assembler scans the assembly language instructions in sequence.
- When it meets a symbolic address, it checks to see if it is already in the symbol table.
- If not, it adds it to the symbol table in the symbolic address column.
- If it is already in the symbol table, check if the absolute address is known.
- If the absolute address is known, it is entered in the appropriate cell.
- If the absolute address is not known, mark/leave as unknown.

Assembler's Second Table

What is the second table that is needed when the assembler software carries out the second pass?

- Op code/ mnemonic/ instruction table

Assembler Pass Actions

- First pass:
  - Creation of symbol table.
  - Expansion of macros.
  - Removal of comments and white spaces.
  - Read the assembly language program one line at a time.
  - Check the opcode in the instruction set.
- Second pass:

    - Read the assembly language program one line at a time.
    - Generation of object code.

Addressing Modes

Addressing Modes

- Direct addressing:
    - The operand is the address where data is stored.
- Indirect addressing:
    - The operand is an address which holds another address where the data is stored.
- Immediate addressing:
    - The operand is the actual value to be loaded.
- Indexed addressing:
    - The contents of the index register are added to the operand to give the actual address.
- Relative addressing
    - The operand is the offset from the current address where the value to be used is stored.

Relative Addressing Purpose

- To allow for relocatable code because all target addresses can be specified by adding the offset to the base address.

Direct vs Indexed Addressing

- Similarity: both load the contents of an address into the accumulator.
- Difference: direct accesses the address given by the operand whereas the indexed adds the contents of IX to the operand and accesses the data at that calculated address.

Assembly Language Instruction Groups

Assembly Language Instruction Groups

- Data Movement
- Arithmetic operations
- Conditional and Unconditional Jump instructions
- Comparison instructions
- Input and output of data
- Shift operations

Instruction Group Descriptions

1. Input and output of data: takes an input from the user and outputs the character of the binary number.
2. Arithmetic operations: performs addition and subtraction.
3. Unconditional and conditional instructions: move to another instruction identified by a label.
4. Compare instructions: compare the result to another value.

Assembly Language Instructions

Data Movement Instruction	Opcode	Instruction	Operand	Explanation
	LDM		#n	Immediate addressing. Load the number n to ACC.
	LDR		#n	Immediate addressing. Load the number n to IX.
	LDD			Direct addressing. Load the contents at the given address to ACC.
	LDI			Indirect addressing. The address to be used is at the given address. Load the contents of this second address to ACC.
	LDX			Indexed addressing. Form the address from *+ the contents of the index register. Copy the contents of this calculated address to ACC.*
	MOV			Move the contents of the accumulator to the given register (IX).
	STO			Store the contents of ACC at the given address.

Arithmetic Operations Instruction	opcode	Instruction	operand	Explanation
	ADD			Add the contents of the given address to the ACC.
	ADD		#n	Add the denary number n to the ACC.

Arithmetic Operations Instruction	opcode	Instruction	operand	Explanation
	SUB			Subtract the contents of the given address from the ACC.
	SUB		#n	Subtract the denary number n from the ACC.
	INC			Add 1 to the contents of the register (ACC or IX).
	DEC			Subtract 1 from the contents of the register (ACC or IX).

Comparisons and Jumps Instruction	Opcode	Instruction	Operand	Explanation
	JMP			Jump to the given address
	CMP			Compare the contents of ACC with the contents of
	CMP		#n	Compare the contents of ACC with the number n
	CMI			Indirect addressing. The address to be used is at the given address. Compare the contents of ACC with the contents of this second address.
	JPE			Following a compare instruction, jump to *if the compare was True.*
	JPN			Following a compare instruction, jump to *if the compare was False.*

Input and Output (no operands)
IN
OUT
END

Key in a character and store its ASCII value in ACC

Output to the screen the character whose ASCII value is stored in ACC

Return control to the operating system

Shift Operations	
LSL	#n
LSR	#n

Bits in the accumulator are shifted logically n places to the left

Bits in the accumulator are shifted logically n places to the right

- Logical shift: where bits in the accumulator are shifted to the right or to the left and zero moves into the bit position vacated.
- Cyclic shift: like a logical shift but bits shifted from one end reappear at the other end.
- Arithmetic shift: uses the shift to carry out multiplications or division of a signed integer stored in the accumulator.
    - Left shift operation is a fast way to multiply by two.
    - Right shift represents integer division by two.
    - Only for unsigned numbers having the most significant bit equal to zero.

Bitwise Logic Operation	
AND	#Bn
AND	<address>
XOR	#Bn
XOR	<address>
OR	#Bn
OR	<address>

Bitwise AND operation of the contents of ACC with the binary number n

Bitwise AND operation of the contents of ACC with the contents of <address>

Bitwise XOR operation of the contents of ACC with the binary number n

Bitwise XOR operation of the contents of ACC with the contents of <address>

Bitwise OR operation of the contents of ACC with the binary number n

Bitwise OR operation of the contents of ACC with the contents of address

Flags in Status Register

Condition Flags

- Zero
- Carry
- Overflow
- Negative

Machine Code Instructions (Extra Notes)

- Machine code consists of a sequence of instructions.
- An instruction contains an opcode.
- An instruction may not have an operand but up to three operands are possible.
- Different processors have different instruction sets associated with them.
- Different processors will have comparable instructions for the same operations, but the coding of the instructions will be different.
- The only language that the CPU recognizes is machine code.
- Therefore, when a program is running and an instruction is

Why use Assembly Language?

Writing a program as a sequence of machine code instructions would be a very time-consuming and error-prone process. Each processor has its own assembly language. For each machine code instruction, there is an equivalent assembly language instruction which comprises:

- A **mnemonic** (symbolic abbreviation) for the **opcode**
- A character representation for the operand

If the program has been written in assembly language it has to be translated into machine code by the assembler before it can be executed by the processor.

Assembler Features

The use of an assembler allows a programmer to include some special features in an assembly language program:

- Comments
- Symbolic names for constants
- Labels for addresses
- Macros

- Subroutines
- Directives
- System calls

The first three items on this list are there to directly assist the programmer in writing the program. Of these, comments are removed by the assembler and symbolic names and labels require a conversion to binary code by the assembler.

Directives and **system calls** are instructions to the assembler as to how it should construct the final executable machine code. They can involve directing how memory should be used or defining files or procedures that will be used. They do not have to be converted to binary.

Assembler Tasks

For any assembler there are a number of things that have to be done with the assembly language code before any translation can be done. Some examples are:

- Removal of comments
- Replacement of a macro name used in an instruction by the list of instructions that constitute the macro definition
- Removal and storage of directives to be acted upon later

Symbolic Addressing

The use of symbolic addressing allows a programmer to write some assembly language code without having to bother about where the code will be stored in memory when the program is run.

Chapter 8: System Software

Operating System

Why does a computer need an operating system?

- Provides software platform/environment on which other programs can be run.
- Provides a user interface and controls communication between user and hardware or hardware and software.
- The hardware is unusable without an OS.

Operating System Tasks

What are the management tasks performed by the operating system?

- **Memory management**

    - Allocates RAM to processes or blocks.
    - Ensures fair usage of memory.
    - Organizes memory.
    - Reclaims unused blocks of RAM.
    - Prevents two processes occupying the same area of RAM at the same time.
    - To release memory when a process stops.
    - Manages paging, segmentation, and virtual memory.
    - Keeps track of allocated and free memory locations.
- **File management**
    - Maintains directory structures.
    - Creates files or folders.
    - Provides file naming conventions.
    - Makes file sharing possible.
    - Space allocated to particular files.
- **Security management**
    - Provides user accounts and passwords.
    - Provide firewalls.
    - Authentication.
    - Implements access rights.
    - Automatic backup.
    - System restore.
- **Hardware/ device/ peripheral management**
    - Install/manage device drivers to allow communication between peripherals and computer.
    - Receives data from input devices and sends data to output devices.
    - Buffer management for transfer of data.
    - Manages interrupts from the device.
    - Control of hardware usage by processes.
    - Device detection.
    - Power management.
    - Keep track of device status.
    - Management of queues.
- Input/ output management.
- Interrupt Handling.
- **Process/Resource management**
    - Manages the scheduling of processes.
    - Allows multi-tasking.

    - Ensures fair access, handles priorities, and handles interrupts.
    - Manages which resources the processes require.
    - Enables processes to share information.
    - Resolution of conflicts.
- **Error checking and recovery**
    - Deals with interrupts.
    - Deals with run time errors generated by software.
    - Deals with hardware faults.
    - Error diagnostic messages.
    - System shutdown.
    - Safe-mode boot-up routines.
- **Provision of a platform for software**
    - Provides an environment within which programs can be run.
- **Provision of a user interface**
    - Allows a user to communicate with the hardware by making navigation around the system easier.
    - Provides facility for user inputting data.
    - Provides facility for outputting to the user.
    - Example: GUI/ command line.

Utility Software

What is a utility program?

> An additional program that helps to maintain or configure the system.

Describe the purpose of utility software in a computer.

- To help users optimize and maintain the computer.
- By making memory allocation more efficient and by checking the system for faults.

Different Utility Programs and Their Purposes

Utility Program	Purpose
Disk formatter	Initializes a disk for first use. Partitions disk into logical drives and sets up the file allocation table. Sets up the file system on the disk.
Disk repair/analysis	Identifies errors and bad sectors so errors can be repaired, corrupted files can be restored and bad sectors are no longer used. Disk contents analysis checks for the errors and disk repair attempts to fix the errors. Reduces access time by optimizing storage.

Utility Program	Purpose
Disk defragmenter	Rearranges individual files to occupy contiguous blocks, improving disk access times by reducing the need to search for the next block of the file. Less head movement is required, and a larger area of free space is created by collecting all free space together as files become contiguous.
Backup	Makes a copy of data at regular intervals on another medium to allow for retrieval in case the original is lost. Allows the user to decide what is backed up and to set up an off-site backup. May encrypt the backup files. Restores the data if necessary.
File compression	Reduces the size of a file, saving storage and memory space, by using algorithms to change the data.
Virus checker	Scans files stored on a computer system for malicious code, as well as scanning files when they enter the system. Sets up a schedule for virus-checking, quarantines viruses, regularly updates the virus definitions, and frees up RAM by removing replicating software.

Program Libraries

What are program libraries?

> Program libraries store pre-written and pre-compiled functions and subroutines.

The program library can be referenced/imported, and the functions and routines can be called in the program to perform common complex tasks.

Advantages of Library Routines

- Saves time as the code is already written.
- Code is already tested so it is more robust.
- Program library routines may perform complex tasks that programmer may not know how to code.
- If there is an improvement in the library routine the program updates automatically.
- Precompiled.
- Makes use of another programmer’s knowledge.
- The library routine code should conform to industry standards and therefore contribute towards a more robust program.

Drawbacks of Library Routines

- Compatibility issues as it may not work with the other code and may require changing program for it to work.

- Not guaranteed through testing as there may be unknown or unexpected bugs.
- Library routine may not meet exact needs so it may give unexpected results.
- If library routine is changed there may be unexpected results.

Dynamic Link Library (DLL)

> A collection of self-contained shared library programs that are already compiled.

It is linked to the main program during execution. Library program code is separate from the .exe file. Library file only loaded into memory when required at run time. A DLL file can be made available to several applications at the same time. If DLL routine is updated the program that uses it will run the update.

Advantages of DLL Files

- Main memory requirements for program are reduced as dynamic link library is loaded when required.
- Executable file size is smaller because the executable does not contain all the library routines.
- No need to recompile the main program when changes are made to DLL because changes to the DLL file code are done independently of the main program.
- A single DLL file can be used in several application programs at the same time saving space in memory.
- Maintenance not needed to be done by the programmer because the DLL is separate from program.
- DLL routines are pre-written saving the developers' time.
- DLL routines are pre-tested so they should be reliable.
- Developers can take advantage of other programmers' expertise.

Disadvantages of DLL Files

- Program will not work if DLL is corrupted.
- An external change to the DLL could stop the program working or change the way it works.
- The DLL file must be present at runtime otherwise there is an error.
- Appropriate software must be available at run-time to import the DLL files.

Language Translators

Why is a Language Translator Used?

To convert a higher-level programming language to a different form.

The 3 Language Translators

- Compiler
- Interpreter
- Assembler

Features of a Compiler

- Converts high level language code into low level code.
- Translates whole program before attempting to run it.
- Creates an executable file if there are no errors.
- Generates a report of all the errors.
- Used after program is complete.
- Used to repeatedly test the same section without having to re-interpret every time.
- Compiler makes it more difficult for the user to modify the code of the program supplied to the user.

Features of an Interpreter

- Converts high level language code into low level code.
- Translates and executes each line of source code one line at a time.
- Must be present in memory to execute the code.
- It stops translation as soon as it finds an error.
- The error is corrected and the program continues running from where it stopped.
- The source code should be present when the user's program is run.
- Used while writing the program to test or debug the partially completed program.

Advantages of a Compiler

- The compiler produces an executable file.
- A program can be distributed without the source code; the user cannot edit the code.
- The compiler is not required to run the program.
- Code does not have to be compiled each time it is run so the program can be tested and run multiple times without recompilation.
- A compiler produces code that executes faster than an interpreted code as it is faster to run the executable file than to interpret the source code every time the program is run.
- Cross-compilation is possible so the program can be compiled to run on different platforms.

Advantages of an Interpreter

- Easier debugging for each stage of program development because it translates line by line and stops when an error is found → one error is reported at a time → errors can be corrected in real-time.
- Parts of the program can be tested without all the program code being available.
- The effect of any change made to the code can be seen immediately → avoid dependent errors.
- Errors are reported as the interpreter finds them.

Drawbacks of a Compiler

- The program will not run if there any errors.
- The program must be recompiled after every change.
- Slower to produce the object code than an interpreter.
- Part-programs cannot be tested.
- One error may result in other false errors being reported.

Drawbacks of an Interpreter

- Source code is needed at run time.
- No executable file produced.
- Execution time increased.

Advantages of Partially Compiled and Partially Interpreted Languages

- Partially compiled programs can be used on different platforms as they are interpreted when run.
- Code is optimized for the CPU as machine code is generated at run time.
- Source code doesn't need recompiling so more efficient to run.

Disadvantages of Partially Compiled and Partially Interpreted Languages

- Intermediate code still needs to be interpreted on the user's computer which may run slowly.
- Extra CPU resources may be required.

Machine Independent

The program code can be translated to run on any processor/platform. The source code is translated into machine-independent intermediate code not machine-dependent code.

Java Translation

Java code uses a two-step translation process and is partially interpreted and partially compiled. The code is translated first into intermediate code using the Java compiler. The bytecode is finally interpreted by the Java Virtual Machine.

Integrated Development Environment (IDE)

Features that Support Visual Presentation

1. **Prettyprinting**
   - Refers to the presentation of the program code typed into an editor. Python IDE for example:
     - Automatically color codes keywords, built-in function calls, comments, strings, and the identifier in a function header.
     - Indentation is automatic.
2. **Expanding and collapsing code blocks**
   - Saves excess scrolling as you can collapse blocks of statements.
3. **Auto indentation/formatting**

Features that Support Writing/Developing the Code

1. **Context-sensitive prompts**
   - As the code is being written, predictions of the code being entered for text completion, such as keywords and available identifiers, are provided.

Features that Support Debugging the Code

1. **Dynamic Syntax Checks**
   - Underlines or highlights syntax errors as the code is entered.
2. **Breakpoint**
   - Running the program and stopping the code at a specific line to check the current values.
3. **Single stepping**
   - Run the code one line at a time so the values can be checked, and the effects of each statement can be seen.
4. **Report window**
   - Outputs the contents of variables and data structures.

Chapter 9: Security, Privacy, and Data Integrity

Meaning of Data Integrity, Privacy, and Security

Describe what is meant by data integrity, data privacy, and data security and the differences between them:

Concept	Description	Examples
Data integrity	Ensures that data is accurate, valid, and up to date. Ensures that data received is the same as the data sent.	Verification and validation, including parity checks, double entry, etc.
Data privacy	Prevents unauthorized access to data. Restricts access to data to keep it confidential and only seen by authorized people.	Restricting access to data to keep it confidential and only seen by authorized people.
Data security	Keeping data safe. Protects data against accidental or malicious loss and from corruption.	Usernames and passwords, firewalls.

Security Measures for Protecting Computer Systems

What are methods to improve the security of computer systems?

Security Measure	Description
Firewall	Prevents unauthorized access to the data. Monitors incoming and outgoing traffic. Checks against a set of rules for acceptable data. Blocks transmissions from unauthorized sources and signals that do not meet requirements. Maintains an allow list and deny list of IP addresses.
User accounts	User has a username and password. Access to resources can be limited to specific accounts. Cannot access system without valid username and password (prevents unauthorized access to the system).
Anti-malware software	Scans for malicious software. Quarantines or deletes any malicious software found. Scans can be scheduled at regular intervals. Should be kept up to date.
Physical measures	Locked doors/keyboards. Secure methods of access, keypads, biometric scans.
Authentication methods	Such as passwords and username to ensure only users with the correct password can access the data. Passwords should be strong → more challenging to guess. Another example is biometrics. Two factor authentication may be used too.

Security Measure	Description
Digital signature	Mathematical algorithm or encrypted data that is attached to electronically transmitted document. Used to verify its content and that it comes from a trusted source.

Digital Signature Authentication

Explain how a digital signature is used to authenticate a digital document during transmission over a network.

1. The sender puts the document through a hashing algorithm to produce a digest.
2. The sender encrypts the digests with the sender's private key to create the digital signature.
3. The message and the signature are sent to the receiver.
4. The receiver decrypts the signature to reproduce the digest with matching sender's public key.
5. The receiver uses the same hashing algorithm on the document received to produce a second digest.
6. The receiver compares this digest with the one from the digital signature.
7. If both of the receiver's digests are the same, the document is authentic.

Security Measures for Protecting Data

What are examples of threats to data that are posed by networks and the internet?

- Malware/viruses/spyware.
- Phishing: sending an email or electronic message from an apparently legitimate source requesting the user to disclose confidential information.
- Pharming: a malicious code that automatically redirects the user to a fake website.
- Hacking

Similarities Between Phishing and Pharming

- Both try to obtain financial or personal information.
- Both make use of fake websites.

Similarities and Differences Between Spyware and Viruses

Feature	Spyware	Virus
Data Access	Only records/accesses data	Can damage computer data

Feature	Spyware	Virus
Data Transmission	Sends the recorded data to a third party	Doesn't send data out of the computer
Replication	Does not replicate itself	Replicates itself
Commonality	Both are pieces of malicious software that are downloaded without the user's knowledge	
Background Operation	Both run in the background	

Threats and Security Measures

Threat	Description	Security Measure
Malware	Malicious software that has the intention of causing harm to a system or its contents such as deleting data.	Install and run up-to-date anti-malware
Virus	A malicious software that replicates and can delete/damage data.	Install and run up-to-date antivirus
Hacker/Unauthorized access	Illegal access with a malicious intent to delete/damage files/data	Use a firewall. Use strong passwords.
Spyware	A malware installed on the computer without the user's knowledge to secretly record keystrokes and sends the gathered data to a third party	Use a firewall. Install and run up-to-date antispyware

Software to Prevent Threats Over a Network

Type of Software	Description
Antivirus/Antispyware/Antimalware	Scans the computer for viruses/spyware and checks against a stored database of viruses, that needs to be updated regularly and then deletes them.
Firewall	Monitors incoming and outgoing traffic and compares it to criteria and blocks those that don't match the criteria.

Methods to Protect Data from Accidental Loss and Unauthorized Access

Method	Description
Access rights	Different access rights for individuals. To stop users editing data they are not permitted to access.
Backup of data	Copies of data are made at regular intervals. They are stored in a different location. If data is corrupted, it can be restored.
Disk mirroring	Data is written to different disks simultaneously. In the event of hardware failure, a second identical copy is immediately available.
Encryption	Data is scrambled using an encryption key into cipher text. Thus, it is meaningless to anyone without a decryption key. Only authorized users will have the decryption key to unscramble the contents.

Factors to Consider When Planning a Backup Procedure

- How often should the data be backed up.
- What medium should the data be backed up to.
- Where should the backups be stored.
- What should be backed up.
- When should the backup take place.
- Who is responsible for performing the backup.

Security Measures for Authorized Employee Access

- Strong passwords.
- Biometrics (e.g., fingerprint recognition).
- Swipe cards.
- Two-factor authentication/Two-step verification.

Importance of Data and System Security

- To stop the data from being amended or corrupted.
- Computer systems need protection to stop people from installing malware or damaging the system.

Restricting Malware Risks

- Install and run an updated anti-malware program so that regular scans can be made for known malware and if found, it will be removed.
- Using a firewall to block traffic that doesn’t meet the criteria → malware can’t enter the computer system.
- Backup so that they can be restored in case of data loss from malware program installation.

Data Validation and Verification

Validation

> Checks that the data entered is reasonable, acceptable, and meets certain criteria. Does not guarantee that data is accurate. Automatic check by computer.

Verification

> Checks that the data entered is the same as the original. Comparison of two versions of data. In the event of a mismatch, the user is forced to re-enter the data. Another definition is: confirmation of data received by system.

Examples of Validation

Validation Type	Description
Length check	Checks that the data is of a specific length, for example, with a telephone number.
Format check	Checks that data is of a certain format, for example, the date has to be dd/mm/yyyy.
Range check	Checks that data is between a certain minimum and maximum, for example, the month in a date should not exceed 12.
Presence check	Makes sure that data must be entered, and the field is not left blank.
Limit check	Ensures that data should be below a certain maximum.
Existence check	A certain file should already exist when referred to in data entry.
Type check	Ensures that data is of the correct data type.

Examples of Verification on Data Entry / Ways of Maintaining Data Integrity at Input Stage

- **Double entry**
  - The data from the form is entered twice by two different people.
  - The two copies of data are then automatically compared by the computer to check they are the same.
- **Visual check**
  - The data is compared manually by two different people after entry to the paper form

where the user compares the input data to the data found in the original document.

Examples of Verification During Data Transfer / Ensuring Data Integrity During Transmission

- **Checksum**
  - The checksum value is calculated from the data before transmission.
  - The result is transmitted with the data.
  - Calculation is repeated at receiving end from the received data.
  - Results are compared.
  - If the checksum received and calculated are different, an error has occurred.
- **Parity check**
  - Uses even or odd parity which is decided before data is sent.
  - Each byte has a parity bit.
  - Counts if the number of 1s or 0s in each byte is odd or even.
  - The parity bit is set to 1 or 0 to make parity for byte correct (odd/even).
  - After transmission, the parity of each byte is re-checked at the receiving end.
  - If it is different, then an error is flagged.

Parity Checks Protect Data Integrity

- It automatically checks for errors on receipt of the data and alerts if data has been received incorrectly.
- It provides a verification check on data.

Check Digit

> When a series of numbers are used to identify something, it is possible to use a check digit method of verification.

Requires a calculation to be made with the numbers that have been entered. The final part of the calculation involves an integer division from which the remainder is added as an extra one or two digits at the end of the series of numbers. This technique is used for a bar code or for the ISBN for a book.

Parity Block Check for Identifying Incorrect Bit Position

- Each byte has a parity bit (horizontal parity).
- An additional parity byte is sent with vertical and horizontal parity.
- Each row and column must have an even/odd number of 1s.
- Identify the incorrect row and column.

- The intersection is the error.

Limitations of Parity Check

- Errors in an even number of bits in the same row or column.
- They prevent the error being identified so it could appear to be correct.

Correct Data After Validation and Verification

Validation checks that data is reasonable, but it doesn't check that accurate data has

Ethics and Ownership

Ethics

> Ethics are a system of **moral principles** that guide behavior or decision-making based on philosophical or religious views. They encompass respectful and considerate behavior.

ACM/IEEE Software Engineering Code of Ethics

A professional **code of conduct** is needed by a company to:

- Identify permissible employee actions.
- Identify prohibited employee actions.
- Identify repercussions for prohibited employee actions.
- Define company and employee responsibilities.
- Define company values and missions.
- Provide a standard for measuring behavior.
- Ensure compliance with legislation.
- Ensure adherence to uniform standards among employees.

Eight Principles of ACM/IEEE Software Engineering Code of Ethics

- **PUBLIC**: Act consistently with the public interest.
- **CLIENT AND EMPLOYER**: Act in the best interests of the client and employer.
    - Involve the client in development.
    - Keep the project on time.
    - Inform the client of any problems or delays.
- **PRODUCT**: Ensure products and modifications meet the highest professional standards possible.
    - Ensure high product standard and requirement fulfillment.
    - Ensure product delivery within time and budget.

    - Ensure well-documented and thoroughly tested product.
- **JUDGEMENT**: Maintain integrity and independence in professional judgment.
- **MANAGEMENT**: Promote an ethical approach to software development and maintenance management.
- **PROFESSION**: Advance the integrity and reputation of the profession.
- **COLLEAGUES**: Be fair to and supportive of colleagues.
    - Work together as a team and listen to each other’s ideas.
    - Assist colleagues in professional development.
    - Fully credit the work of other team members.
    - Make colleagues aware of expected standard ways of working.
- **SELF**: Participate in lifelong learning and promote an ethical approach to the profession.

Importance of Signing a Professional Code of Conduct

A new software engineer is asked to sign a professional code of conduct to:

- Confirm understanding of the code of conduct.
- Ensure awareness of expected behavior and consequences of prohibited actions.
- Ensure adherence to uniform standards among employees.

Importance of Joining a Professional Ethical Body

It's important for a programmer to join a professional ethical body because:

- The programmer has ethical guidelines to follow, removing the need to decide what is ethical.
- Clients gain confidence in the programmer's reputation and skills.
- The ethical body provides help and support, such as legal advice.

Ownership and Copyright

> **Copyright** refers to the formal and legal rights to ownership/intellectual property rights, protecting against unauthorized reproduction of work and providing a legal right of redress.

Purposes of Copyrighting

- Protects intellectual property rights by placing formal restrictions on the use of the program.
- Allows legal action against anyone else who claims it as their own.
- Formally identify the client as the owner of the software.

Software Licensing

Types of Software Licensing

Licensing Type	Description
Commercial	A license must be purchased to use the software, which defines how it can be used as the program is copyrighted. Source code is not distributed, restricting modifications and preventing illegal changes. It also restricts the number of users, possible time period for use, and limited installations are allowed as a software key is needed to install.
Shareware	Users get a trial period for free, which may be limited in features. After the expiry date, users must pay to continue using the software and unlock additional features and get full functionality. Users can redistribute the software, but do not have access to the source code.
Free software	Users can edit and redistribute the source code.
Open-source software	The source code is distributed with the software, allowing users to modify and edit it. It is usually free of cost, but users must redistribute modified software under the same terms.
Freeware	There is no charge for the software, but it could still be copyrighted. Owners can set restrictions on what a user can do with the program, and modification and re-distribution of the software without permission is prohibited.

Benefits of Commercial Licenses

To the owner:

- The owner can charge a fee for the software.
- The owner retains the copyright, preventing redistribution without permission.
- Commercial license prohibits any changes to the software, meaning the user can't legally edit the program.
- Legal action can be taken against people who attempt to illegally copy it.

To the client:

- Potentially better support as the owner is charging a fee.
- Likely to have fewer bugs and be less prone to malware than software distributed under other licenses.
- Redress available if the software does not function correctly.

Benefits of Shareware Licenses

To the client:

- They can check if it works or meets their requirements without having to pay a fee if it does not.
- They have peace of mind that the software hasn't been maliciously edited.

To the programmer:

- Users can try the program and offer constructive feedback.
- More people might try the program because the trial is free of charge.
- Users who find the trial useful will buy the program so the programmer gets income.
- Allows the program to be copyrighted and so protects the programmer's intellectual property rights.

Advantages of Distributing Programs Under an Open-Source License

- The user can customize the code.
- Errors can be identified, reported, and/or fixed by the users.
- Additional features can be added to the code.

Artificial Intelligence

> **Artificial Intelligence (AI)** concerns the use of a computer or computer-controlled device to perform tasks normally associated with intelligent behavior by humans.

How AI Plays a Board Game

- The rules or past moves of the game will be stored.
- The AI program is trained by playing many times.
- AI will look ahead at possible moves and analyse the pattern of past choices.
- Then it chooses the move most likely to be successful.
- Computers could learn how to improve or learn from previous mistakes by storing the positive and negative result of choices.
- It then changes its future choices.

Applications of AI

- **Police identifying wanted people**: it uses image recognition to identify features in an image.
- **Natural language interfaces**: it uses speech recognition to identify words that are spoken and adapts to learn regional accents.

- **Self-driving car**: it detects its position on the road and within the traffic to follow a route, avoid a collision, or self-park.

Social Impact of Facial Recognition Software

- **Incorrect recognition of faces** can lead to mistakes, such as denial of access.
- **Privacy issues** arise as people do not like data being stored.
- **Individuals will be safer** as there might be a reduction in crime.
- **Faster boarding**.
- **Catching criminals**.

Economic Impact of AI in a Repair Garage

- Costs of the garage are reduced because less time is taken for diagnosis.
- Profits for the garage increase as technicians spend more time repairing, leading to more jobs completed in a day.
- Costs passed to the customer are decreased, potentially attracting more customers.

AI in CCTV Systems

- Uses image recognition.
- Monitors every image taken and tries to identify the key features that match a ‘person’.
- Starts recording to secondary storage when a person is identified.
- The system identifies the direction of movement of a person and uses this to decide where to move the camera.

AI in Speech-to-Command Conversion

- Uses speech recognition.
- Which identifies the key phrases that were spoken and matches these to a database.
- Then, it generates the most likely sentence.

AI in Camera Autofocus

- Uses facial recognition.
- Scans the scene in real time.
- Looks for patterns that match the pattern of a face and identifies if there are faces in the image.
- Camera focuses on the pattern identified.

Chapter 10 Extra Notes

The Public Good

In the code, there is reference to:

- Health, safety, and welfare of the public
- Public interest
- Public good
- Public concern

Examples include:

- Ariane 5 rocket
- Powerful companies exerting pressure on less powerful companies.
- Companies not providing security against unauthorized access to their systems.
- Social media sites allowing abusive content to be transmitted.
- Search engines providing search results with no concern about the quality.

Ownership and Copyright

> **Ownership** is when an individual/organization is the owner if it creates and publishes some work that has an element of originality.

Copyright can be applied to:

- Literary work
- Music composition
- Film
- Music recording
- Work of art
- Computer program

How to Avoid Abuses of Copyright

- Legislation is introduced for such a reason.
- A copyright legislation will include:
  - Requirement for registration, recording date of creation of work
  - Defined period when copyright will apply
  - A policy to be applied if an individual holding copyright dies
  - An agreed method for indicating the copyright, for example, the use of the symbol ©

Software Licensing

> The special feature in free software is called "**copyleft**," which is the condition that if the software is modified, the source code for the modified version must be made available to other users under the same conditions of usage.

Artificial Intelligence

AI depends on and draws from many other disciplines including philosophy, psychology, neuroscience, mathematics.

5 Aspects of Intelligent Human Behavior and AI Applications

- **Problem-solving**
  - Chess: Rules are limited so a computer with sufficient storage and processing can investigate so many options for possible sequence of moves that human cannot.
  - Traditional form of expert system: for example medical aid
    - Supplied by data and rules from living medical experts.
    - System has more knowledge than human can store.
    - If given a new situation that is not covered by rules given earlier, no creative approach is attempted.
- **Linguistics**
  - Voice recognition and voice synthesis techniques.
  - Example is if you phone a helpline, and a computer processes your sound to transfer you the appropriate human.
- **Perception**
  - Robots are used in the manufacturing process to perform repetitive tasks.
  - If anything unexpected happens the robot continues to operate as normal regardless of any damage caused
  - Research now is focused on autonomous robots to be fitted with sensors to enable the robot to take appropriate action depending on information received.
- **Reasoning**
  - AI applications have been able to draw inferences (reach conclusions based on evidence) which is a requirement for reasoning.
  - Example is providing mathematical theorems.
- **Learning**
  - This area is concerned with machine learning.
  - Machine learning is said to take place if the system that performs a task improves its performance as it gains experience.
  - AI system has access to experience in the form of massive set of data and by using mathematical statistical algorithms, the system learns from this data.

    - Example: if actions of users visiting the website to buy products is stored, when the user revisits the AI system, identify appropriate products to advertise (if sales increase then there's evidence that learning is taking place)

Impact of AI

- Organizations providing systems that collect and store user activity of how the internet is being used
    - If data is being used to increase profits, then this is considered normal business practice
    - If not stored securely, it could get into the hands of wrong people and used for criminal activity
- Introduction of autonomous mechanical products such as robots, robotic devices, and driverless vehicles.

Advantages and Disadvantages of AI

Advantages	Disadvantages
Arguments that technological developments lead to employment of more people to manufacture, install, and service.	Another argument is that this leads to fewer jobs as machines are doing the work.
People are excited for driverless vehicles.	The potential of accidents will increase as there is not enough measures to prevent accidents.
Robots can be used in environments that are dangerous for humans to enter.	Robots require materials for their construction.
	All electronic/mechanical devices end up causing serious problems of waste products harming the environment.

Chapter 11: Databases

Limitations of a File-based Approach

The disadvantages of a file-based approach compared to a relational database are:

- **Data duplication or more data redundancy**: The same data is stored multiple times unnecessarily in different files.
- **Data inconsistency and reduced data integrity**: Duplicated data might be stored as different values, and when data is updated in one place, it is not updated everywhere.

- **Program-data dependency**: If the data structure changes, all the programs accessing the data must be changed and re-written too.
- **Lack of data privacy**: No access control to file parts, and user views cannot be easily implemented.

Benefits of Relational Databases

The benefits of using a relational database instead of a file-based database are:

- **Linked tables can be set up**: Different tables can be linked by common attributes.
- **Reduced data redundancy**: Linked tables mean that each item of data is only stored once.
- **Improved data integrity**: Changes in one table will automatically update in another, and linked data can't be entered differently in two tables.
- **Provide different views and privacy is improved**: Different users have different access to the data of the database.
- **Allows program-data independence**.

Overcoming File-Based Disadvantages

A relational database overcomes the disadvantages of the file-based approach by:

- **Multiple tables are linked together**: Which reduces data redundancy or duplication.
- **Increases data integrity (consistency)**: Reduces compatibility issues; data only needs to be updated once, and associated data will be automatically updated.
- **Program-data independence**: The structure of data can change and does not affect the program, and the structure of programs can change and does not affect data; the data can be accessed by any appropriate program.
- **Different users can be given different access rights and different views of data**: Improves security and maintains data privacy.
- **Referential integrity can be enforced**: Unwanted or accidental deletion of linked data is prevented; users cannot enter data for a record that does not exist; users cannot delete a record while it still exists in another table.
- **Complex queries can be more easily written**: They search specific data.

Reducing Data Redundancy

Data redundancy is reduced in relational databases by:

- Each record is stored once and is referenced by a primary key.
- Data is stored in individual tables, and the tables are linked by relationships.
- Proper use of primary and foreign keys.
- Enforcing referential integrity.

- Going through the normalization process.

The Database Approach

The three levels of the schema of a database are:

- **External**: The individual's view of the database.
- **Conceptual**: Describes the data as seen by the applications making use of the DBMS; describes the views which users of the database might have.
- **Internal (physical)**: Describes how the data will be stored on the physical media.

> A **logical schema** is an overview of a database structure that models the problem by using methods such as an ER diagram. It is independent of any particular DBMS.

DBMS Features

> A **data dictionary** is a file/table that stores all the information about the database and metadata about the database.

Data Items Stored in a Data Dictionary

- Table names: to identify the tables.
- Field names or attributes
- Data types: to prevent the wrong data type from being entered in a field.
- Type of validation
- Primary keys
- Foreign keys: to link tables and create relationships between them.
- Relationships

A table can have a composite primary key because neither key uniquely identifies each tuple by itself.

Tasks Performed Using the DBMS Developer Interface

- Create a table.
- Set up relationships between tables.
- Create a form.
- Design a report to output data.
- Create a query.
- Create interactive features such as buttons.
- Create a menu to select options for different actions.
- Add objects to a form to make data input easier such as drop-down boxes.

> A **query processor** is software that is used to create, process, and execute SQL queries. It allows the user to enter criteria and searches and returns data that meets set criteria and organizes the results to be displayed to the user; performs calculations on extracted data.

An **index** for a table is used because the database user will frequently want to search for a specific field. That specific field or attribute will be indexed. It allows for a faster search using that attribute. If a table contains a lot of data, it is quicker to search in an index table than in the full table.

Security Features Provided by the DBMS

- **Access rights**: Restrict actions (e.g. read/read-write) of specific users can do on parts of the database; unauthorized users cannot access the database; give managers access to different elements by having different accounts which have different access rights (e.g. read/read-write).
- **Views**: Different users are able to seed certain parts of the database.
- **Usernames and passwords or biometrics**: Prevents unauthorized access.
- **Automatic backup**: Automatically create regular copies of the database and store off-site on a regular basis in case of loss.
- **Encryption**: Data is turned into ciphertext so that unauthorized users can't be understood without a decryption key.

The Relational Database

Term	Definition
Entity	An object about which data can be stored; the concept or object in the system that we want to model and store information about.
Table/Relation	All the data about one entity organized in rows and columns where columns form the structure and the rows form the content.
Field	A column or attribute in a table.
Attribute	A data item represented as a column or field in a table.
Record/Tuple	The data in one row of a table about one instance of an object.
Primary key	Attribute or combination of attributes that is used to uniquely identify a record.
Foreign key	A field in one table that links to the primary key in another table.
Candidate key	Any attribute or combination of attributes that can be a primary key.
Secondary key	A candidate key that has not been chosen as the primary key.

A relationship between two tables is implemented by the primary key in one table linking to the foreign key in another table.

Entity Relationship Modelling

Relationship Type	Example
One-to-one	
Many-to-one and one-to-many	Relationship between a player and team: One team has many players; player can only play in one team.
Many-to-many	Relationship between team and match as many teams can play in the same matches and many matches are played by the same team; relationship between members of a club and classes they take.

Logical Entity Relationship Model

- M:M cannot be implemented on a relational database as the foreign key attribute can only have a single value so it cannot handle the many references required
- To solve this, a link entity is created between those 2 entities
- The M:M relationship is changed into 2 M:1 or 1:M relationships which can be handled by the relational database
- In the team, match example it will be represented like this:
  - Team
  - Match
  - Team-Match
- There will now be 2 foreign keys in the link entity, one referencing the primary key in the team table and the other referencing the primary key in the match table.

Referential Integrity

> **Referential integrity** ensures related data in tables are consistent and up to date; ensures that every foreign key value has a corresponding value in the corresponding primary key; making sure tables do not try to reference data that does not exist (a value of one attribute of a table exists as a value of another attribute in a different table); make sure that if data is changed in one place, the change is reflected in all related records.

A primary key cannot be updated/deleted unless all dependent records in the foreign tables are already updated (cascading update). The foreign keys must be the same data type as the corresponding primary key.

> **Data integrity** refers to methods of making sure the data is consistent.

Examples include enforcing referential integrity and validation/verification rules.

Normalisation

Normalization Forms (NFs)

- **1NF (1st Normal Form)**
  - No repeated groups of attributes.
  - There must be a unique primary key.
  - All attributes should be atomic.
  - No duplicate rows.
- **2NF (2nd Normal Form)**
  - No partial dependencies (non-key attributes should be dependent on the compound key, not just one of the keys).
  - Should be in 1NF.
- **3NF (3rd Normal Form)**
  - No non-key dependencies (no non-key attributes are dependent on other non-key attributes) → no transitive dependencies.
  - All fields are fully dependent on the primary key and no other attributes.
  - No many-to-many relationships.
  - Should be in 1NF and 2NF → no repeated groups of attributes and no partial key dependencies.

> A **repeating group** is a set of attributes that have more than one set of values when the other attributes each have a single value.

SQL (Structured Query Language)

An industry standard language that provides both DML and DDL statements.

General SQL Rules

- Consists of sequence of commands.
- Each command is terminated by ;
- Command can occupy more than one line.
- No case sensitivity, yet we write the commands in upper case and table names, attributes in lower case.
- Command that contains a list will be separated by comma.

1. Data Definition Language (DDL)

DDL can be used to create:

- Database
- Tables
- Fields/attributes
- Indexes
- Users
- Primary key
- Foreign key
- Relationships
- Views

Other Tasks Performed Using DDL

Used for altering tables by altering something within the structure of the table (example adding a new attribute to existing table), not the values in the table.

SQL Commands Representing Tasks Performed by DDL

```sql
-- Create a new table with specified columns and data types
CREATE TABLE TableName (
    ColumnName DataType
);

-- Example: Create a Band table with BandName as primary key
CREATE TABLE Band (
    BandName VARCHAR(25),
    PRIMARY KEY (BandName)
);
-- VARCHAR is a variable-length text data type. (25) is the max number of
characters.

-- ALTER (Modify table structure)

-- Add a new column to an existing table
ALTER TABLE Band
ADD NumberOfMembers INTEGER;

-- Add a primary key to an existing table
ALTER TABLE Band
ADD PRIMARY KEY (BandName);

-- Add a foreign key referencing another table
ALTER TABLE BandBooking
ADD FOREIGN KEY (BandName) REFERENCES Band(BandName);
```

```sql
-- Remove a column from a table
ALTER TABLE Band
DROP COLUMN NumberOfMembers;
```

Data Manipulation Language (DML)

> DML is used to populate, modify, delete, and read data from tables.

SQL Commands Representing Tasks Performed by DML

```sql
-- INSERT (Add a new row to a table)

-- If the order of attributes is known (less recommended)
INSERT INTO TableName VALUES ('Attribute1Value', 'Attribute2Value');

-- If the order of attributes is unknown (recommended)
INSERT INTO TableName (AttributeName1, AttributeName2, AttributeName3)
VALUES ('value1', 'value2', 'value3');

-- UPDATE (Modify existing data)
UPDATE TableName
SET Attribute = 'NewValue'
WHERE CriteriaAttribute = 'SomeValue';

-- DELETE (Remove rows based on a condition)
DELETE FROM TableName
WHERE CriteriaAttribute = 'CriteriaValue';

-- SELECT (Retrieve data from the table)

-- Get specific data that meets a condition
SELECT NumberOfMembers
FROM Band
WHERE BandName = 'Coldplay';

-- Get all band names in ascending order
SELECT BandName
FROM Band
ORDER BY BandName;

-- For descending order
SELECT BandName
FROM Band
```

```sql
ORDER BY BandName DESC;

-- Show each band name only once
SELECT DISTINCT BandName
FROM Band;

-- Join two tables and filter
SELECT VenueName, Date
FROM Booking
JOIN BandBooking ON BandBooking.BookingID = Booking.BookingID
WHERE BandBooking.BandName = 'Coldplay';

-- Get all data from a table
SELECT *
FROM Band;

-- Get average number of members
SELECT AVG(NumberOfMembers)
FROM Band;

-- Get total number of members
SELECT SUM(NumberOfMembers)
FROM Band;

-- Count all rows in the table
SELECT COUNT(*)
FROM Band;

-- Count rows where NumberOfMembers is not null
SELECT COUNT(NumberOfMembers)
FROM Band;
```